

1. Newton's first law of motion is also called the law of:

- a) inertia
- b) acceleration
- c) action and reaction
- d) gravitation

Answer: a

2. According to Newton's second law, force equals:

- a) mass times acceleration
- b) mass divided by acceleration
- c) acceleration divided by mass
- d) mass plus acceleration

Answer: a

3. What is the SI unit of force?

- a) Joule
- b) Newton
- c) Watt
- d) Pascal

Answer: b

4. Linear momentum is defined as:

- a) mass times velocity
- b) mass times acceleration
- c) force times distance
- d) velocity divided by time

Answer: a

5. Work done by a constant force is:

- a) $F \times d \times \cos \theta$
- b) F / d
- c) $m \times g$
- d) $F \times t$

Answer: a

6. What is the SI unit of energy?

- a) Newton
- b) Joule
- c) Pascal
- d) Hertz

Answer: b

7. The acceleration due to gravity on Earth is approximately:

- a) 9.8 m/s^2
- b) 98 m/s^2
- c) 0.98 m/s^2
- d) 980 m/s^2

Answer: a

8. The energy possessed by a moving body is called:

- a) potential energy
- b) kinetic energy
- c) thermal energy
- d) nuclear energy

Answer: b

9. In projectile motion, the horizontal velocity is:

- a) uniform
- b) accelerated
- c) random
- d) zero

Answer: a

10. Power is defined as:

- a) work divided by time
- b) work multiplied by time
- c) force times distance
- d) mass times velocity

Answer: a

11. State Newton's three laws of motion.

(Write your answer in the space provided.)

12. Explain the principle of conservation of linear momentum.

(Write your answer in the space provided.)